Ex. No. : 04 **Date:**

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A PYTHON PROGRAM TO IMPLEMENT SINGLE LAYER PERCEPTRON

Aim:

To implement a **Single Layer Perceptron model in Python** to classify **Iris dataset**.

Requirements:

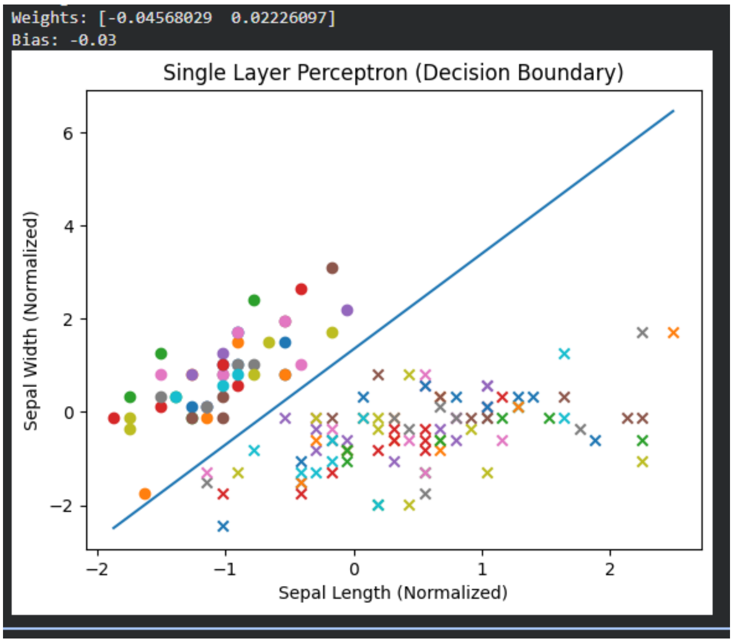
Jupyter Notebook with Python, Iris.csv dataset, and required Python libraries.

PROGRAM:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

uploaded = files.upload()

df = pd.read_csv('Iris.csv')

df['target'] = df['Species'].apply(lambda x: 1 if x == 'Iris-setosa' else -1)

X = df[['SepalLengthCm', 'SepalWidthCm']].values
Y = df['target'].values

X = (X - X.mean(axis=0)) / X.std(axis=0)

w = np.zeros(2)
b = 0
lr = 0.01
epochs = 100

for _ in range(epochs):
    for i in range(len(X)):
        if Y[i] * (np.dot(w, X[i]) + b) <= 0:
            w += lr * Y[i] * X[i
```

```
b += lr * Y[i]

print("Weights:", w)
print("Bias:", b)

plt.figure()

for i in range(len(X)):
    if Y[i] == 1:
        plt.scatter(X[i,0], X[i,1])
    else:
        plt.scatter(X[i,0], X[i,1], marker='x')

x_vals = np.linspace(X[:,0].min(), X[:,0].max(), 100)
y_vals = -(w[0]*x_vals + b) / w[1]

plt.plot(x_vals, y_vals)

plt.title("Single Layer Perceptron (Decision Boundary)")
plt.xlabel("Sepal Length (Normalized)")
plt.ylabel("Sepal Width (Normalized)")

plt.show()
```

OUTPUT:



Result:

Thus, the Single Layer Perceptron model was successfully implemented using Python and evaluated using the Diabetes dataset.